AMBULANCE AND EMERGENCY DISPATCH SYSTEM

Project Architecture & System Workflow

1. Project Overview

The Ambulance and Emergency Dispatch System aims to reduce emergency response time by prioritizing emergency requests and assigning a suitable available ambulance based on urgency and distance.

2. Project Objectives

- Prioritize critical emergency requests.
- Identify the nearest available ambulances.
- Calculate shortest routes using Dijkstra's algorithm.
- Prevent duplicate ambulance assignments.
- Check hospital distance and bed availability.

3. System Workflow

1. Receive an emergency request.
2. Validate request details and assign emergency priority.
3. Identify available ambulances.
4. Calculate routes and select a suitable ambulance.
5. Select a hospital based on distance and bed availability.
6. Display the dispatch result and assignment status.

4. Technologies Used

C++ • Data Structures and Algorithms • Priority Queue / Max-Heap • Dijkstra's Algorithm • MySQL • GitHub

5. Team Members

Neel Hajariwala — Team Lead
Keshav Namdev
Lichhavi Chauhan
Aditya Ahlawat

Suggested GitHub commit message: Plan project architecture and define system workflow

Note: This PDF documents the project overview and workflow. Upload it as a file only if it reflects the actual work in your repository; a GitHub commit is created when you commit a real file change.